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Acceptance in Principle (AIP)

Once peer review is completed and the editor is satisfied that any concerns raised by the reviewers have been suitably addressed, the paper is accepted 'in principle' (AIP). The editor will send detailed instructions to ensure that the paper contains all the information required for publication and meets our standards for the reporting of data and code.

A high priority of *Nature Machine Intelligence* is that all papers are accessible to non-specialists; manuscripts are subject to substantial editing to achieve this goal. After acceptance, a copy editor and an art editor may make further changes to ensure that the text and figures are readable and clear to those outside the field, and that papers conform to our house style.

Once your manuscript has been accepted in principle for publication, you will need to format it according to the guidelines below. Addressing these requirements at this stage will ensure your manuscript has a smoother path to publication.

FORMATTING REQUIREMENTS

1. Document type
2. Tables
3. Figures
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5. References
6. Methods
7. Acknowledgements
8. Funding Statements
9. Extended data figures
10. Source data
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13. Gene nomenclature
14. Chemical and biological nomenclature and abbreviations
15. Equations
16. Cover artwork

1 – Document type

Please submit your manuscript in either Word or TeX/LaTeX format. We do not accept PDFs for final submissions.

Word

We accept all standard fonts; however, please note Symbol font should be used for Greek characters.

TeX/LaTeX

- To submit a TeX/LaTeX file, please use any of the standard class files such as article.cls, revtex.cls or amsart.cls.

- All textual material should be provided as a single file in default Computer Modern fonts.

- Please avoid non-standard fonts and packages and remove all personal macros before submitting.

- For graphics, we recommend graphicx.sty.

- Please use numerical references only for citations, and include the references within the manuscript file itself.

- If you wish to use BibTeX, please copy the reference list from the .bbl file, paste it into the main manuscript .tex file, and delete the associated \bibliography and \bibliographystyle commands.

- Before submission, please ensure that the complete .tex file compiles successfully on your own system with no errors or warnings.

- There is no need to spend time visually formatting the manuscript: our style will be imposed automatically when the paper is prepared for publication.

For further guidelines on TeX/LaTeX, please visit https://www.springernature.com/gp/authors/campaigns/latex-author-support where you can download the Springer Nature

LaTeX template.

2 – Tables

- Please include tables at the end of your text document.
- Complex tables can be submitted as a separate Excel file.
- Tables that include statistical analysis of data should describe their standards of error analysis and ranges in a table legend.
- Tables that feature chemical structures should be included at the end of the text document and the native ChemDraw file(s) should be supplied separately as .cdx files.

3 – Figures

Effective figure preparation will enhance the readability of your manuscript – our image preparation guidelines will help you to produce effective publication-quality figures. In addition, please consider the following important requirements:

Images

- All figures must be cited in sequence within the main article text in the form Fig. 1, Fig. 2.
- Figure panels should be prepared at a minimum resolution of 300 dpi and saved at a maximum width of 180 mm.
- Use a 5–7 pt san serif font for standard text labelling and

Symbol font for Greek characters.

- Use scale bars, not magnification factors, and include error bars where appropriate.

- Do not flatten labelling or scale/error bars onto images – uneditable or low-resolution images are two of the most common reasons for delay in manuscript preparation.

Figure legends

- Include a brief title for each figure with a short description of each panel cited in sequence.

- Ensure the legend does not exceed the word limit of the article type.

- Avoid methodological detail.

- Use verbal cues to describe keys, eg. "open red triangles", not visual cues or symbols.

- Include a description of centre values (median or average) and all error bars and how they were calculated. Give an indication of sample size (n number), state the statistical test used and provide P values.

Chemical structures

Chemical structures should be produced using ChemDraw or a

similar program. Figures containing chemical structures should be submitted according to our image preparation guidelines. Authors using ChemDraw should make use of our journal template, submitting the final files at 100% as .cdx files. Creating molecules within or copying them into the template will ensure that most of our journal style points are followed. For more information, please see our Chemical Structures Style Guide.

Stereo images

Stereo diagrams should be presented for divergent 'wall-eyed' viewing, with the two panels separated by ~5.5 cm. In the final accepted version of the manuscript, stereo images should be submitted at their anticipated final size.

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Advance Online Publication

Nature Machine Intelligence provides Advance Online Publication (AOP) of research articles, which benefits authors

with an earlier publication date and allows our readers access to published papers before they appear in a journal issue. Note that papers published online are definitive and may be altered only through the publication of a formal correction, so authors should make every effort to ensure that they check proofs carefully. All AOP articles are given a unique digital object identifier (DOI) number, which can be used to cite the paper before issue publication. Follow this link for details about advance online publication.

4 – Colour charges (print journals only)

Charges apply for authors of primary subscription content who wish to publish figures in colour in Nature Portfolio print journals. For further details please email editorial enquiries.

5 – References

References should be numbered sequentially in the following order:

1. Main Text
2. Methods
3. Data Availability Section
4. Tables
5. Figure Legends
6. Box
7. Extended Data Figures

References that are uncited or cited out of sequence may require manual reworking in production that could delay the

production process. Please ensure the following:

- Only one publication is given for each reference.
- Research objects (such as conference abstracts, numbered patents, datasets, protocols and code) that have been assigned a DOI are included in the reference list.
- Unpublished meeting abstracts, papers in preparation and papers under review or in press without an available preprint should not appear in the reference list. Instead, they should be mentioned in the text with a list of authors (or initials if any of the authors are co-authors of the present contribution).
- URLs for web sites should be cited parenthetically in the main text, not in the reference list.
- Grant details and acknowledgments are not permitted as numbered references.
- Footnotes are not supported.

For long-form manuscripts such as Articles, Letters, Review Articles, Perspectives and Progress Articles, titles of cited articles are required.

Example:

Eigler, D. M. & Schweizer, E. K. Positioning single atoms with a

scanning tunnelling microscope. *Nature* 344, 524–526 (1990).

For shorter article types such as Commentaries or News & Views, titles of cited articles are not included.

Example:

Iijima, S. *Nature* 354, 56–58 (1991).

For book citations, the publisher is required.

Example:

Jones, R. A. L. *Soft Machines: Nanotechnology and Life* Ch. 3 (Oxford Univ. Press, 2004).

Data and code may be cited in the reference list if they have been assigned DOIs and should include authors, title, publisher (repository name) and DOI expressed as a URL.

Example:

Hao, Z., AghaKouchak, A., Nakhjiri, N. & Farahmand, A. Global Integrated Drought Monitoring and Prediction System (GIDMaPS) data sets. figshare http://dx.doi.org/10.6084/m9.figshare.853801 (2014).

To cite a preprint

Example:

Babichev, S. A., Ries, J. & Lvovsky, A. I. Quantum scissors: teleportation of single-mode optical states by means of a nonlocal single photon. Preprint at https://arXiv.org/quant-ph/0208066 (2002).

6 – Methods

Methods should be subdivided by short, bold headings referring to methods used. We encourage the inclusion of specific subsections; for example Statistics, Reagents or Animal models. If additional references are included, numbering should continue from the end of the last reference in the main paper (see section 5 References). Any Methods-only references should accompany the additional Methods at the end of the paper.

Authors are encouraged to deposit the step-by-step protocols used in their study to Protocol Exchange. Protocols deposited by the authors should be mentioned in the Methods section and added to the reference list.

7 – Acknowledgements

Acknowledgements should be brief and avoid thanking anonymous referees and editors. Grant or contribution numbers may be acknowledged. Dedications are not permitted unless for someone directly involved with the research who for some reason is not an author.

8 – Funding Statements

Any relevant funding should be declared in a separate funding statement. When declaring funding, our recommended best

practice is that authors should note funders and grants on publications when the activities that contributed to that publication:

- are within scope of the grant and
- arise directly from a specific grant

An appropriate format for a funding statement is:

A.B.C. discloses support for the research of this work from Funder [grant number xxxx], Funder [grant number xxxx] and Funder [grant number xxxx]. D.E.F. discloses support for publication of this work from Funder [grant number xxxx], Funder [grant number xxxx] and Funder [grant number xxxx]. G.H.I. declares no relevant funding.

For more information see the full Nature Portfolio Funding Statements policy page.

9 – Extended data figures

Extended Data figures are display items that provide essential background to the data presented in the main paper.

- A maximum of 10 Extended Data display figures is permitted.

- Extended Data figures are not copy-edited or styled by *Nature Machine Intelligence*; for this reason, please follow journal style as closely as possible when preparing these figures.

- Figures can be multi-paneled, if necessary, and should be sized such that they will fit on a single PDF page.

- Legends for Extended Data figures should be prepared as for main figures and should be included in the 'Inventory of Supporting Information' document.

- All Extended Data figures must be referred to as discrete items at an appropriate place in the main text.

10 – Source Data

We encourage you to provide source data for your figures whenever possible. Full-length, unprocessed gels and blots must be provided as source data for any relevant figures, and should be provided as individual PDF files for each figure containing all supporting blots and/or gels with the linked figure noted directly in the file. Statistics source data should be provided in Excel format, one file for each relevant figure, with the linked figure noted directly in the file. For imaging source data, we encourage deposition to a relevant repository due to size constraints.

11 – Supplementary information

Supplementary information is published as supplied to *Nature Machine Intelligence*. Please ensure that it is presented clearly and succinctly in a logical order, and that terminology conforms with the rest of the paper. Once your paper is in production, Supplementary Information can only be replaced to correct significant scientific errors.

Requirements for Supplementary Information

- Each item should be designated as either Supplementary Equation, Discussion, Notes, Figure, Table, Video, Audio, Data or Software, and all except Notes should be numbered sequentially. This numbering should be separate from that used for items appearing either in the main article or in the Extended Data.

- Supplementary Figures should be used only for cases when the use of Extended Data to report these findings is not appropriate. Each Supplementary Figure should fit, along with its legend, on a single PDF page. Authors are encouraged to be selective in including other types of supplementary information.

- Supplementary Figures, Tables and Videos should have a title and a caption that follow the guidelines for display items in the main text.

- We encourage all methodological information to be included in the main text but if additional information is needed, e.g. for algorithm description, step-by-step protocol, compound synthesis and characterization, they can be included as a Supplementary Note.

- Refer to each piece of supplementary information at least once within the text of either the main article or its Methods, at the appropriate point(s). Be sure to include the word "Supplementary" each time one is mentioned and cite

them in sequence.

Please submit supplementary text, figures, simple tables or data, and associated legends within a single combined PDF file. For complex tables or data (larger than an A4 PDF page) we also accept tables and data as Excel workbooks or .csv files named 'Supplementary Tables' or 'Supplementary Data' as appropriate. We accept most tables and data as .csv files. We accept most commonly used audio and video formats, and supplementary software should be submitted within a .zip or .tar archive file.

12 – New structures

Manuscripts reporting new structures should contain a table summarizing structural and refinement statistics. Templates for such tables describing cryo-EM, NMR and X-ray crystallography data are available. To facilitate assessment of the quality of the structural data, a stereo image of a portion of the electron density map (for crystallography papers) or of the superimposed lowest energy structures (>10; for NMR papers) should be provided with the submitted manuscript. If the reported structure represents a novel overall fold, a stereo image of the entire structure (as a backbone trace) should also be provided. For cryo-EM structures, a representative micrograph showing individual particles should be provided in the submission.

13 – Gene nomenclature

Authors should use approved nomenclature for gene symbols, and use symbols rather than italicized full names (Ttn, not titin).

Please consult the appropriate nomenclature databases for correct gene names and symbols. A useful resource is Entrez Gene. Approved human gene symbols are provided by HUGO Gene Nomenclature Committee (HGNC), e-mail: hgnc@genenames.org; see also http://www.genenames.org. Approved mouse symbols are provided by The Jackson Laboratory, e-mail: nomen@informatics.jax.org; see also www.informatics.jax.org/mgihome/nomen.

For proposed gene names that are not already approved, please submit the gene symbols to the appropriate nomenclature committees as soon as possible, as these must be deposited and approved before publication of an article.

Avoid listing multiple names of genes (or proteins) separated by a slash, as in 'Oct4/Pou5f1', as this is ambiguous (it could mean a ratio, a complex, alternative names or different subunits). Use one name throughout and include the other at first mention: 'Oct4 (also known as Pou5f1)'.

14 – Chemical and biological nomenclature and abbreviations

When possible, authors should refer to chemical compounds and biomolecules using systematic nomenclature, preferably using IUPAC and IUBMB rules. Standard chemical and biological abbreviations should be used. Unconventional or specialist abbreviations should be defined at their first occurrence in the text.

15 – Equations

Equations and mathematical expressions should be provided in the main text of the paper. Equations that are referred to in the text are identified by parenthetical numbers, such as (1), and are referred to in the manuscript as "equation (1)".

16 – Cover artwork

Authors of accepted papers are encouraged to submit images related to their research for consideration as an issue cover once their paper is assigned to an issue. Cover images are selected for their scientific interest and aesthetic appeal. Please upload good quality image files along with a clear and concise legend explaining the image content.

Images created with Generative AI (such as from DALL-E, Stable Diffusion, and Midjourney) are not permitted. View additional cover guidance in the Nature guide.

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